

Hygiene Investment Urgency Index

A data-driven WASH investment priority tool
for policymakers at the World Bank, UNICEF, and WHO

Shanqi (Christine) · Xinran · Hannah

BUS32120 · Final Project



The Decision Problem

Where should WASH funding go?

87

countries

×

4

WASH interventions

= 348 funding choices

Today's reality

Allocation decisions are often made without integrated, cross-source WASH evidence.

Funding may not flow to the countries — or the interventions — where the health return is greatest.

Our goal: replace guesswork with an evidence-based priority ranking.

Our Approach

Five stages — from raw data to a ranked priority list



Data Sources & SQL Pipeline

Five complementary sources, merged into one country-year table

Sources

- DHS Program API**
Household WASH & child health outcomes
- World Bank (core)**
GDP per capita, life expectancy, population
- World Bank (literacy)**
Female literacy rate
- World Bank (health)**
Health expenditure %, urbanization
- OECD**
Official development assistance (ODA)

Merge

LEFT JOIN on iso3 + year

Each row = one country in one survey year. Joins preserve all DHS rows; macro columns fill in where available.

371

country-year
observations

87

countries

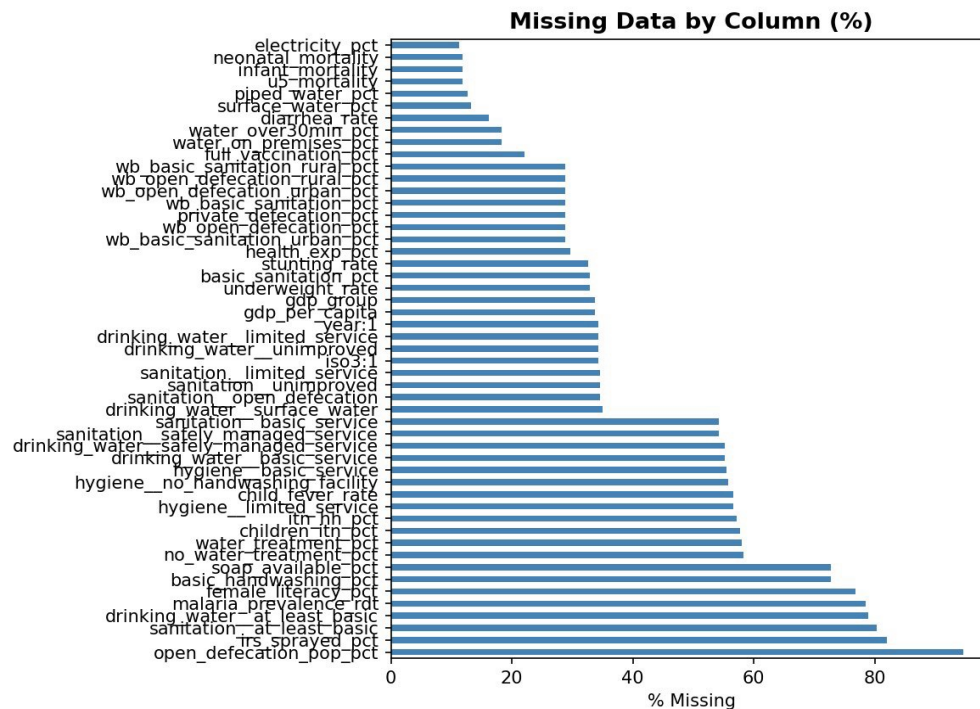
40

years
(1985–2024)

Why SQL: reproducible joins, transparent logic, easy to audit by anyone reviewing the analysis.

Data Quality: What You Should Know

Honest caveats before we get into the analysis



COVERAGE

Uneven by country

DHS rounds vary — some countries appear 27 times, others only once.

MISSINGNESS

Structural, not random

Gaps reflect which DHS modules were fielded — not measurement error.

EXTREMES

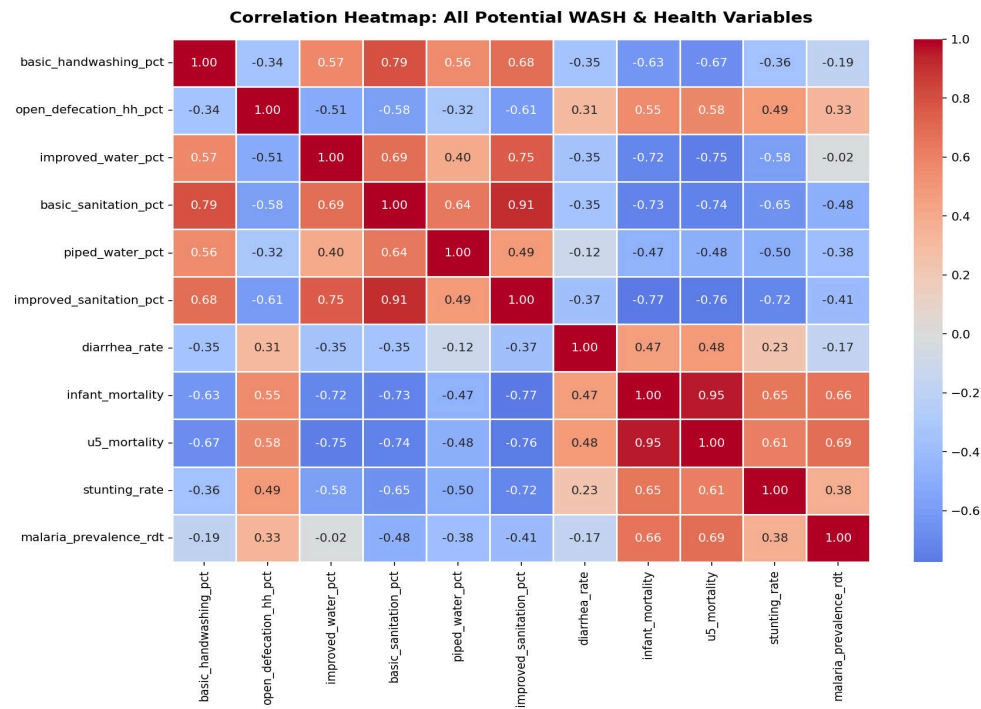
Real, not anomalies

e.g.: Niger 1992: u5 mortality = 322 per 1,000 (real conditions).

Implication: trust regional patterns more than country-specific values where DHS coverage is thin.

What the Data Tells Us

Correlation heatmap: WASH indicators vs. health outcomes



Open-defecation

$r = 0.58$

with under-5 mortality

Basic sanitation

$r = -0.76$

with under-5 mortality

Improved water

$r = -0.75$

with under-5 mortality

Basic handwashing

$r = -0.67$

with under-5 mortality

Modeling Approach

Quantify how much each WASH dimension actually moves child mortality

Models

Linear regression × 3

Predicts each continuous outcome:

- diarrhea rate
- infant mortality
- **under-5 mortality** ← weights for the Index

Logistic regression × 1

Binary: high-risk if infant mortality > median.

Robustness check — 81.2% accuracy.

309

country-year observations
after regional-mean imputation

Why standardize features?

Variables operate on different scales (GDP in thousands; WASH % bounded 0–100). Standardizing to mean=0, std=1 makes coefficients directly comparable, using them as weights in the Urgency Index.

Regression Results

Standardized coefficients — which WASH lever moves mortality most?

WASH Variable	Diarrhea rate	Infant mortality	Under-5 mortality ★
Improved water %	−1.06	−10.31	−21.93
Open defecation (HH) %	+0.96	+3.60	+7.75
Basic handwashing %	−1.74	+0.21	−5.60
Basic sanitation %	+0.27	−3.10	−2.52
<i>Model R²</i>	<i>0.159</i>	<i>0.583</i>	<i>0.634</i>

★ U-5 model has the highest R² and is used as weights for the Urgency Index.

KEY INSIGHT

Improved water

|−21.9|

is roughly

3× larger

than the next biggest WASH lever

Each unit of standardized water access change is associated with ~22 fewer deaths per 1,000 live births.

Building the Urgency Index

Combining health impact with a country's room for improvement

$$\text{Urgency}_i = \Sigma (\text{Gap}_i \times |\beta_i|)$$

GAP

Room for improvement

Distance from full coverage in each WASH dimension, normalized 0–1.

Example: a country at 20% water access has a much larger gap than one at 90%.

COEFFICIENT $|\beta|$

Health impact per unit

From the under-5 mortality regression ($R^2 = 0.63$ — best-fitting model).

Tells us how much child mortality moves per unit change in that dimension.

Final index rescaled to 0–100 — where 100 = highest investment urgency.

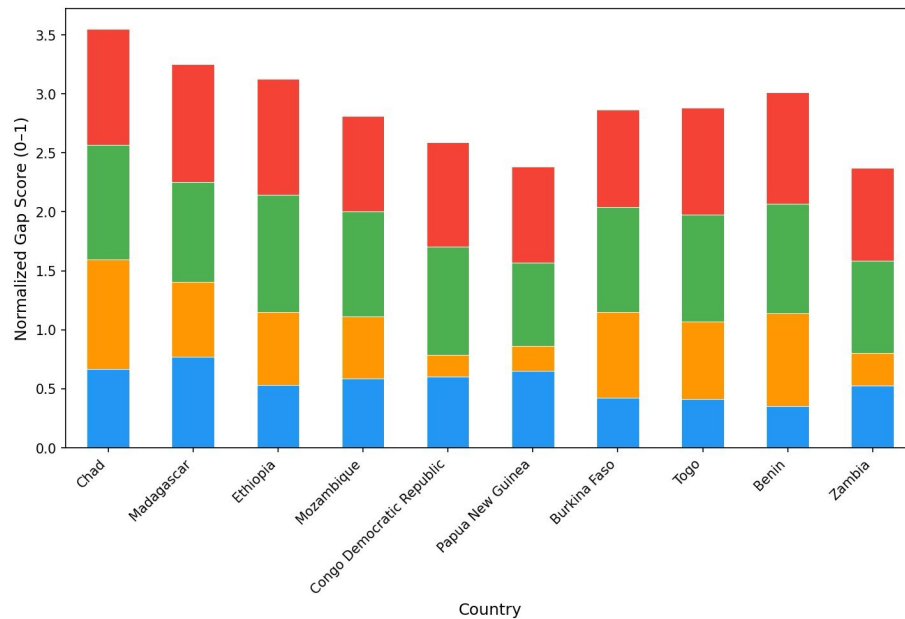
Where to Invest, Within Each Country

Adding the regression weights changes which dimension matters most



RAW GAP — by size only

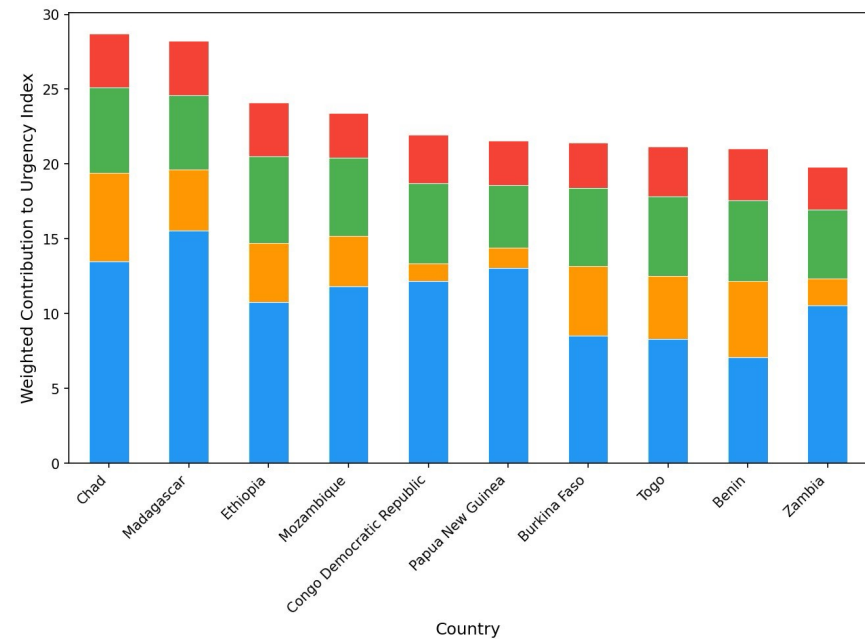
WASH Gaps for Top 10 Highest Urgency Countries (Unweighted)



Sanitation gap looks dominant

WEIGHTED — by health impact

Weighted WASH Investment Gaps — Top 10 Highest Urgency Countries



Water dominates — biggest mortality lever

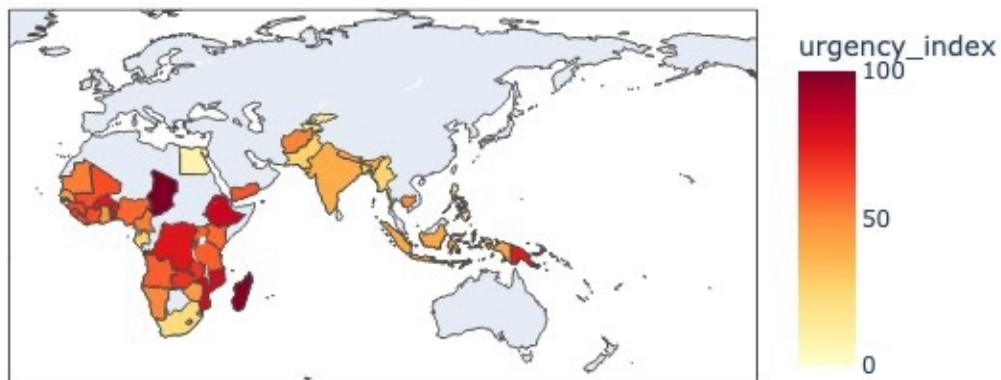
Implication: a country's biggest raw gap isn't always its highest-return intervention. Weighting reveals that water comes first.

Where the Need Concentrates

Urgency Index by country — Sub-Saharan Africa dominates

TOP 10 HIGHEST URGENCY

WASH Investment Urgency Index by Country



1	Chad	SSA	100.0
2	Madagascar	SSA	98.4
3	Ethiopia	SSA	83.8
4	Mozambique	SSA	81.3
5	Congo DR	SSA	76.2
6	Papua New Guinea	OCE	74.8
7	Burkina Faso	SSA	74.4
8	Togo	SSA	73.4
9	Benin	SSA	73.0
10	Zambia	SSA	68.7

9 of the top 10 are in Sub-Saharan Africa; Papua New Guinea is the outlier.

Recommendations for WB · UNICEF · WHO

Three concrete priorities from the index

1

GEOGRAPHIC PRIORITY

Concentrate on SSA + PNG

9 of 10 highest-urgency countries are in Sub-Saharan Africa. Oceania has one hotspot — Papua New Guinea.

2

INTERVENTION
PRIORITY

Water first, within every country

Improved water has the largest mortality coefficient (~3× the next lever). Highest return per dollar.

3

INDEX IS VALIDATED

Higher urgency = higher real deaths

High-tier countries (n=17) average 129 deaths/1,000 — vs. 53 in Low. The index tracks reality.

Validation: average actual under-5 mortality by index tier

129

HIGH (n=17)

91

MEDIUM (n=23)

53

LOW (n=13)

Questions?